

Year 4 Science Revision: Chapters 1 to 4

Student Name: _____ Date: _____

Teacher: IGCSE Y4 Science

Total Marks: / 50

Section A: Multiple Choice Questions (15 Marks)

Circle or highlight the correct answer for each question.

1. What are the three main functions of the human skeleton?
A) Pumping blood, breathing, thinking C) Eating, sleeping, growing
B) Support, protection, movement D) Making food, storing water, running
2. Which of the following is an adaptation for an animal in a snowy habitat?
A) Large, thin ears to release heat C) Thick white fur and blubber
B) Webbed feet for swimming D) A long sticky tongue
3. In a food chain, a plant is always a:
A) Consumer B) Predator C) Prey D) Producer
4. When a muscle contracts to move a bone, what happens to it?
A) It gets longer and thinner C) It turns into a bone
B) It gets shorter and thicker D) It stays the exact same size
5. Which of the following statements about solids is true?
A) They can be easily squashed. tern.
B) They take the shape of the container. D) They can flow and be poured.
C) Particles are packed tightly in a pat-
6. What is the process called when a liquid turns into a solid?
A) Melting B) Freezing C) Evaporation D) Condensation
7. Which of the following is a **natural** source of light?
A) A torch B) The Sun C) A mirror D) A glass window
8. How is a shadow formed?
A) When a transparent object lets light pass C) When an opaque object blocks light
B) When light bends around a corner D) When the sun goes down at night
9. Why do we get vaccines?
A) To cure a headache C) To help bones grow faster
B) To stop infectious diseases D) To digest food properly

10. Animals with a hard outer shell instead of bones have an:
A) Endoskeleton B) Exoskeleton C) Internal spine D) Organ shell
11. Which material allows almost all light to pass through it?
A) Wood B) Cardboard C) Glass D) Metal
12. Which of these is an example of an irreversible change?
A) Melting ice B) Baking a cake C) Freezing water D) Dissolving salt
13. What connects muscles to bones?
A) Joints B) Organs C) Tendons D) Skin
14. What do the arrows in a food chain represent?
A) Which animal is larger C) The direction the wind blows
B) The transfer of energy D) Where the animals live
15. What is the habitat of a camel?
A) Ocean B) Rainforest C) Arctic D) Desert

Section B: Fill in the Blanks (15 Marks)

Word Bank

habitat
energy
skull

joints
relax
opaque

liquid
fair
transparent

friction
medicine
bones

gas
reflect
predator

Choose the correct word from the word bank above to complete each sentence.

1. The environment in which a plant or animal naturally lives is called its _____.
2. All living things need _____ to survive, which animals get from their food.
3. The bones in our head that protect our brain are called the _____.
4. Places where two bones meet, allowing us to bend and move, are called _____.
5. Muscles work in pairs; when one muscle contracts, the other must _____.
6. A material that does not let any light pass through it is called _____.

7. A _____ is a state of matter that can flow and takes the shape of its container.
8. When conducting an experiment, keeping all conditions the same except for the one you are testing makes it a _____ test.
9. Clear glass is an example of a _____ material because light passes clearly through it.
10. A force that slows down moving objects when they rub against each other is _____.
11. A drug prescribed by a doctor to help cure an illness is called _____.
12. The human skeleton is made up of over 200 _____.
13. A state of matter where particles are spread very far apart and move freely is a _____.
14. Mirrors and shiny surfaces are able to _____ light back into our eyes.
15. An animal that hunts and eats other animals is known as a _____.

Section C: Structured Questions (20 Marks)

Write your answers clearly on the lines provided. There are 15 sub-questions in total.

Question 1: The Human Body & Medicines (4 Marks)

a) Name one major organ that the skull protects.

b) Name two major organs that the ribcage protects.

c) Explain how the biceps and triceps work together to bend your arm at the elbow.

d) What is an exoskeleton? Give one example of an animal that has one.

Question 2: Ecosystems and Food Chains (4 Marks)

Look at the food chain below:

Grass → **Grasshopper** → **Frog** → **Snake**

a) Which organism is the **producer** in this food chain?

b) Name one **predator** from the food chain above.

c) What does the arrow (→) in a food chain represent?

d) If a disease killed all the frogs in this habitat, what would likely happen to the population of grasshoppers? Explain your answer.

Question 3: States of Matter (3 Marks)

a) In the box below, draw a simple model showing how particles are arranged in a **Solid**.



b) A student places a solid block of chocolate in the hot sun. What change of state will happen to the chocolate?

c) Is baking a cake an example of a reversible or irreversible change? Why?

Question 4: Light and Shadows (4 Marks)

Sarah is using a torch to shine light on a wooden block. A shadow forms on the wall behind the block.

a) Why does the wooden block form a shadow? Use the word 'opaque' in your answer.

b) Sarah wants to make the shadow on the wall **bigger**. What should she do with the wooden block?

c) If Sarah replaces the wooden block with a clear glass sheet, what will happen to the shadow?

d) Does light travel in straight lines or wavy lines?
